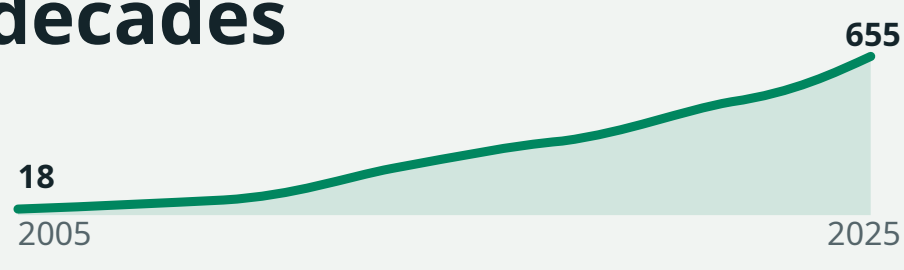


Perturbation evidence is multiplying faster than it can be organised

A single pooled genome-scale Perturb-Seq dataset can **simultaneously profile tens of millions of cells across hundreds of thousands of perturbations**. That scale could reveal robust biology across studies, but identifiers, metadata, assays and processing pipelines differ. The Perturbation Catalogue aims to unify datasets across modalities and provide a shared access path for this evidence.

36× growth in perturbation papers over two decades



Europe PMC title/abstract query; 24 Aug 2026.
Scale example: [FirstWord Pharma, story 7139212](#).

Our approach turns scattered datasets into a unified resource

1,237

datasets

19,235

targets

36

tissues

34

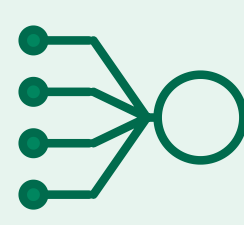
cell types

1,200

cell lines

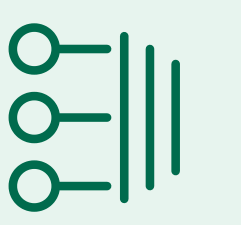
169

diseases



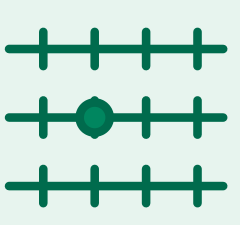
CRISPR screens · 1,201 datasets

Pooled guides → gene-level phenotypes and fitness.



Perturb-Seq · 26 datasets

Guide identity + scRNA-seq → cell-state responses.



MAVE · 10 datasets

Variant libraries + assays → sequence-function maps.

We extensively curate metadata...

84

meta-data fields

Study & provenance

Title, authors, year, URI, licence

Model system

Organism, tissue, cell line, disease

Perturbation & library

Targets, guides, library, variants

Readout & processing

Assay, technology, kit, reference

Stable Ensembl gene IDs + ontology terms make cross-study questions searchable.

And reprocess data from scratch

SRA / FASTQ

Raw sequencing reads and study-specific probe lists

→ **Cell × gene counts**

Shared reference and gene annotation

→ **Guide assignment**

Map cells to perturbations

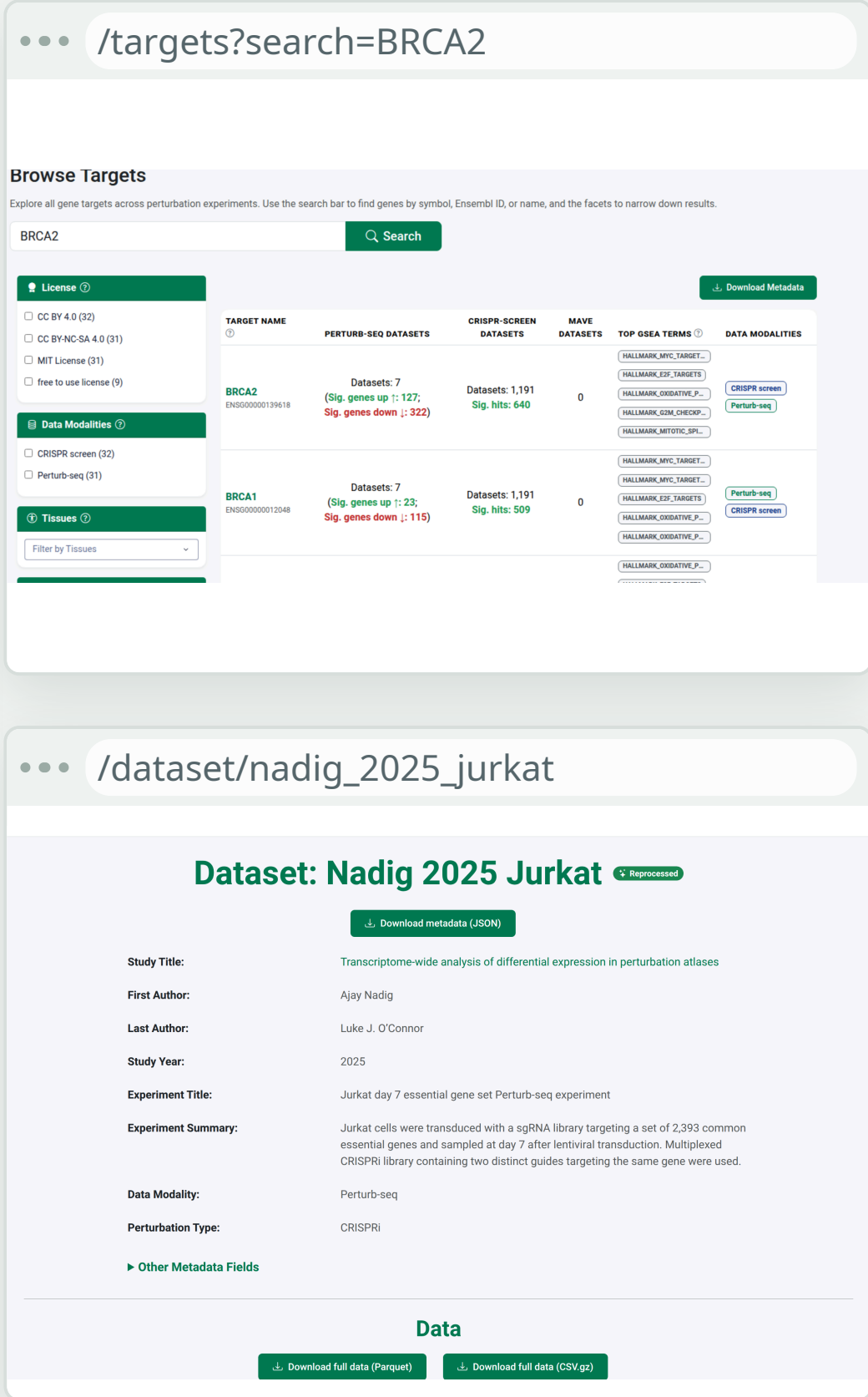
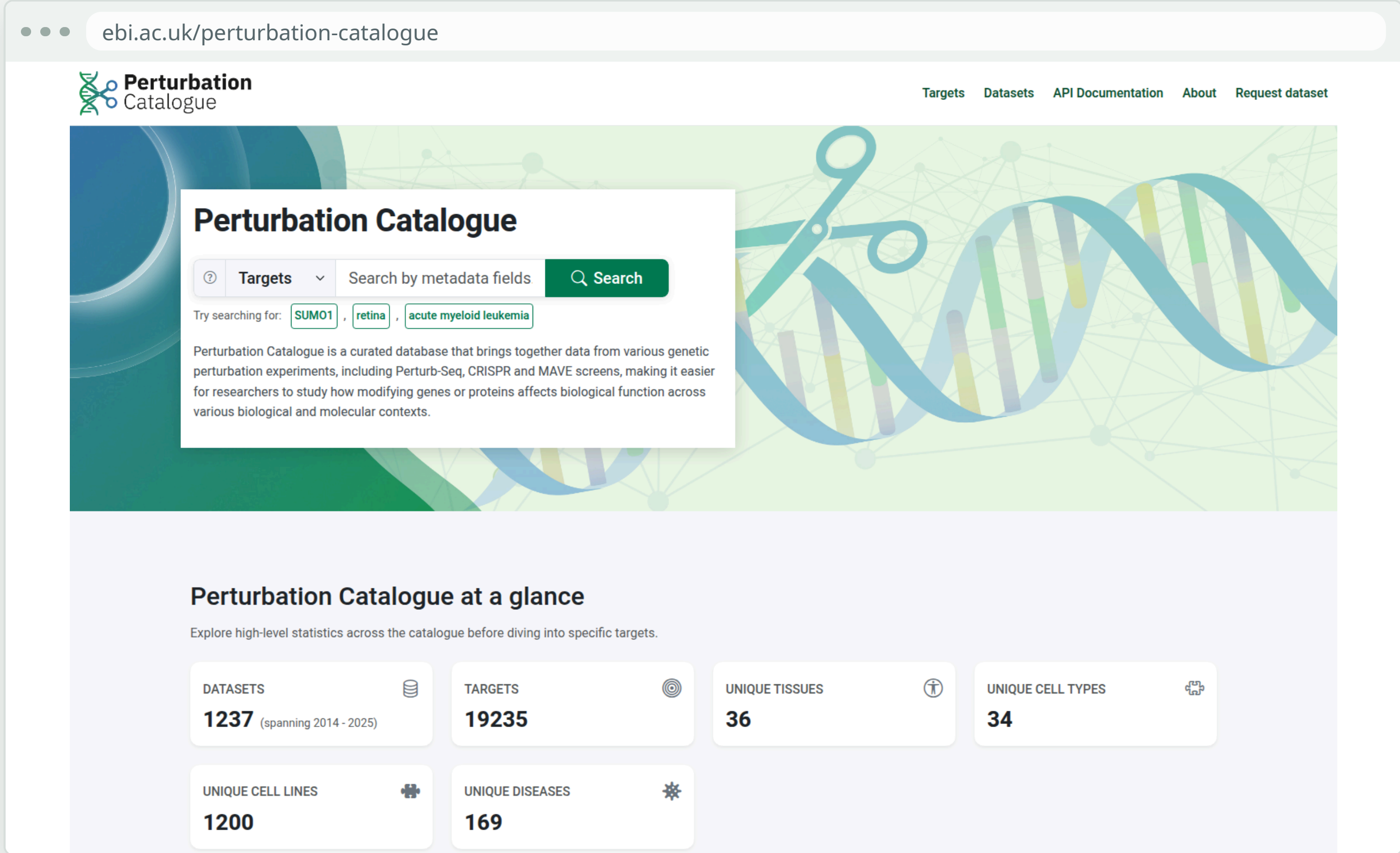
→ **QC + filtering**

Produce the final analysis-ready H5AD

Five Perturb-Seq datasets are live. Perturbation-control comparisons feed DEA and Hallmark GSEA. More datasets and modalities coming soon.

Browse the Catalogue

Search → Context → Filter → Explore



Access records via API

Structured JSON for custom pipelines, reproducible analysis and machine learning.

```
GET /v1/perturb-seq/nadig_2025_jurkat/search?perturbation_gene_name=TFAM&limit=1
```

```
{  "perturbation": {"gene_symbol": "TFAM"},  "effect": {"gene_symbol": "MT-ND5", "log2fc": -1.926, "padj": 0.0}}
```

Download entire datasets

Data is available in **Parquet** and **CSV.gz**, and metadata as **JSON**.

Stay tuned for more exciting things coming soon



Extract structured metadata faster

AI-assisted drafting with curator review and validated ontology mapping.



Reanalyse more designs

More kits, experimental structures and independent quality metrics.



Publish regular, versioned releases

Explicit provenance and stable snapshots for reproducible downstream use.



Reanalyse MAVE data

Enrich variant-to-function coverage with harmonised scores.